

Spotted Python

Week 4 Tutorial

- ★ Simple Graphing
- ★ For and While loops
- ★ Creating subplots

Simple Graphing

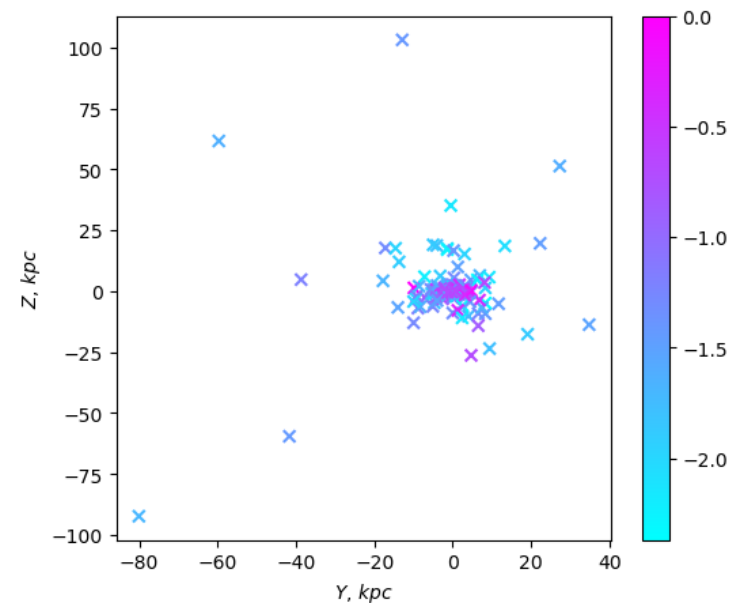
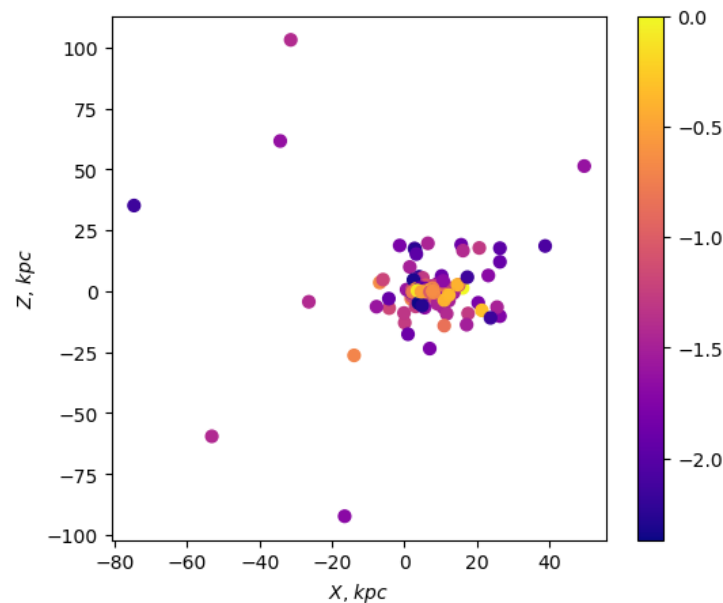
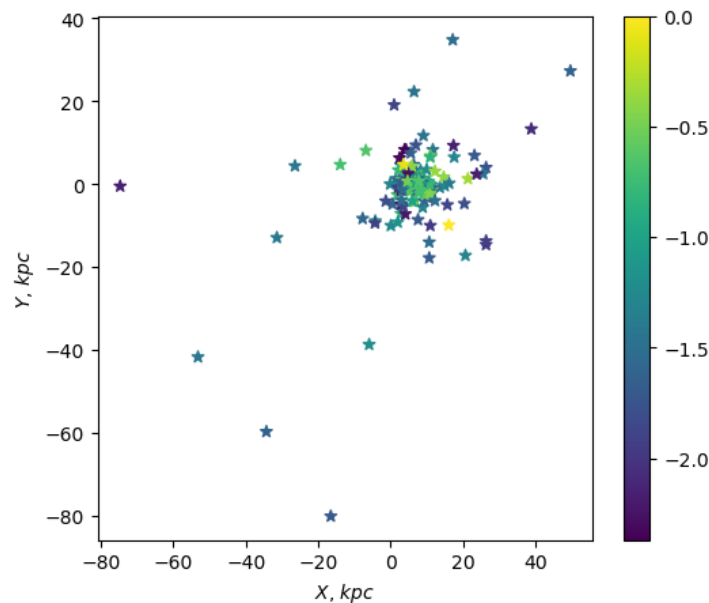
- ★ Matplotlib is the most used tool for visualisation
- ★ Documentation: <https://matplotlib.org/stable/>
- ★ List of in-built colours:
https://matplotlib.org/stable/gallery/color/named_colors.html
- ★ List of in-built colourmaps:
<https://matplotlib.org/stable/users/explain/colors/colormaps.html>
- ★ Module to import alternative colourmaps:
<https://cmasher.readthedocs.io/index.html>
- ★ Tool to create colourmap: <https://gka.github.io/palettes>

For and While Loops

- ★ For loops repeat a piece of code while changing an input variable
- ★ While loops repeat a piece a code until a certain condition is met
- ★ Complete **Exercise 1**

Milky Way Globular Clusters

- ★ Reminder that the assignment data can be found under [assessment hub](#) → [computational assessment data](#)
- ★ For the working example, use the `harris_combined.csv` provided
- ★ Colouring the GC by $[Fe/H]$, what trends do we see?



SAMI

- ★ Relationship between V/σ and ϵ (Cappellari, 2007)

$$\left(\frac{V}{\sigma}\right)_e = \sqrt{\frac{(0.09 + 0.1\epsilon)\epsilon}{1 - \epsilon}}$$

- ★ Try to recreate this figure from van de Sande, 2017 (**Exercise 2**)

